

Computer Network Reference Models

Module 04

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana in this lecture we will study about TCP IP reference model first of all please see the contents which I shall cover after covering brief introduction about TCP IP reference model I shall take up its layered protocol architecture which will be followed by detailed explanation about protocols in tcpip reference model 1 minute, 1 second the Internet Protocol suit commonly known as TCP IP reference model is the computer networking model and set of communication protocols used on the internet it is commonly known as TCP IP because its most important protocols the

transmission control protocol TCP and the Internet Protocol protocol IP were the first networking protocols defined in this standard the TCP IP reference model is the network model used in the current Internet architecture it has its origin back in the late 1960s with the grandfather of the internet the arpanet this was was a research Network sponsored by the Department of Defense in the United States and due to this reason it was also called DOD model the major design goals of tcpip reference model were ability to connect multiple networks together seamlessly ability for connections to remain intact as long as the source and destination machines are functioning to be built on flexible architecture the TCP IP model and related protocol models are maintained

by the internet engineering task force ietf the tcpip internet Pro protocol suit resulted from research and development conducted by defense Advanced research project agency DARPA in the late 1960s after initiating the pioneering arpanet in 1969 DARPA started work on a number of other data transmission technologies in 1972 Robert E KH joined the DARPA information processing technology office where he worked on both satellite packet networks and groundbased radio packet networks and recognized the value of being able to communicate across both in the spring of 1973 vnon serve the developer of the existing arpanet Network control program NCP protocol joined KH to work on open architecture interconnection models with

the goal of Designing the next protocol generation for the arpanet by the summer of 1973 KH and surf had worked out a fundamental reformulation in which the differences between Network protocols were hidden by using a common inter Network protocol and instead of the network being responsible for reliability as in the arpanet the host became responsible the initial four computers connected to the network in the basic arpanet increased to several hundred in just a short time by linking universities research institutes and military installations with each other over Leed telephone lines as technologically different networks such as satellite Network and wireless networks were connected to the internet the originally implement ented protocols were soon overburdened by the data traffic

translation required from one network to the other now let us see TCP IP reference model the actual TCP IP reference model consist of four layers that is layer 2 to 5 as shown in figure these together together with the addition of the 6 minutes, 1 second physical layer that is layer one makes up the five layer hybrid TCP IP reference model lower most layer in TCP IP reference model is link layer which is also known as host to network layer above link layer is internet layer followed by transport layer topmost layer is application layer in tcpip reference model the link layer or host to network layer of the TCP IP reference model corresponds to the first two layers of the iso OSI reference model that is physical layer and data link layer its main task focuses on on

the secure transmission of data packets of pulled bit sequences it is followed by the internet layer which corresponds to the network layer of the isoi reference model its main responsibility is to enable the data communication of Two end systems at a given location in the hetrogeneous communication ation Network the transport layer above it corresponds to the layer of the same name in the iso OSI reference model it enables two user programs on different computers in the communication Network to exchange reliable and connection oriented data the application layer of the TCP IP reference model Model includes the three upper layers of the iso OSI reference model and serves as an interface for the

actual application programs that wish to communicate with each other over the network unlike the iso OSI reference model it was not conceived and planned theoretically but was derived from the protocols that had been put into practice on the internet the isoi protocols on the other hand were planned theoretically and adopted before protocols could be invented that implemented the different functions for the layers of the isoi reference model today isoi Protocols are no longer used and the tcpip reference model protocols which had been developed from practical application dominate the internet layered Communication in TCP IP reference model is as shown in figure here lower two layers link and internet are chained layers and upper two layers

transport and application are end to endend layers where process to process communication takes place now we will see the task and the protocols of the individual layers of the TCP IP reference model first is link layer or Network to host layer the link layer is the lowest layer of

the TCP/IP reference model the link layer defines the networking methods within the scope of the local network link on which hosts communicate without intervening routers. The primary task of the link layer is to secure your transmission of individual data packets between two adjacent end systems. This layer of the TCP/IP reference model is normally subdivided into two further layers: medium access control (MAC) sublayer and logical link control (LLC) sublayer.

The MAC sublayer regulates access to the shared channels by defining a mechanism of allowing fair and efficient access to all participants. This includes methods for the discovery of collisions or their avoidance as many participants wish to transmit data at the same time. The LLC sublayer forms the data link layer of the LAN. It provides flow control and link management. Data transmission errors must be recognized and, if possible, corrected. The data is also subdivided into size-limited data packets as per requirement. In this layer, important protocols of the link layer of the TCP/IP reference model are based on the IEEE 802 LAN standard. These are technologies such as Ethernet, IEEE 802.3 token ring, IEEE 802.5 as well as different wireless LAN technologies like IEEE 802.11 and many more. The most important protocols of the link layer of the TCP/IP protocol family are ATM and synchronous transfer mode (STM).

ATM, which stands for asynchronous transfer mode, is a connection-oriented packet-switching network protocol that breaks down and forwards the data to be transported in cells of a fixed size. Behind this design principle was the idea of providing time-critical real-time data such as video or audio information together with regular data over a standardized protocol. ARP (address resolution protocol) is used within the help of the ARP protocol described in RFC 826. The MAC address of a host can be

determined from the IP address in the Internet Protocol of the layer above it. This is important if a data packet from the Internet is to be delivered to a local network whose MAC address must be determined from the stored IP address of the receiver for forwarding. In the LAN, NDP (neighbor discovery protocol) functions of the NDP protocols are very similar to the ARP protocol and serve to explore and discover further hosts in the local network. In contrast to ARP, NDP was developed for the next generation of the Internet Protocol, IPv6, while ARP works under the current version, IPv4. In the link layer topology discovery, the proprietary LLDP (link layer discovery protocol) was developed by the Microsoft company for exploration.

of the present network topology and verification of the guaranteed quality of service in a network. PPP (Point-to-Point Protocol) is a simple protocol that serves the connection between two network nodes. PPP is used by the majority of Internet providers to offer their customers a dialup connection over a standard telephone line. In the Internet second layer in the TCP/IP model is the Internet layer. The Internet layer has the responsibility of sending packets across potentially multiple networks. It requires sending data from the source network to the destination network. The Internet Protocol performs three basic functions: host addressing and

identification which is accomplished with a hierarchical IP addressing system packet routing which is the basic

task of sending packets of data from source to destination by forwarding them to the next Network router closer to the Final Destination congestion control which is used for control of traffic congestion now let us see two basic protocols at internet layer the center Cal protocol of the internet layer is the Internet Protocol IP IP offers an unreliable data packet oriented endtoend information transmission it is responsible for fragmentation and defragmentation into IP data grams IP exist today in two versions ipv4 and ipv 6 in addition the protocol icmp internet control message protocol is implemented in the internet layer it is responsible for the notification of specific errors occurring during IP

transmission as well as for further diagnos G nostic task such as sending Eco request to test the availability of a computer and the necessary transmission time icmp is a protocol that sits directly on the IP two different variations of the icmp protocol exist one for ipv4 and another for ipv 6 beside IP and icmp there are two more protocols that work on the internet layer of TCP IP protocol stack for example IP SEC Internet Protocol security it is comprised of a protocol suit to ensure the secure execution of Ip data traffic within a data stream IP datagrams can be authenticated and encrypted additionally included in IPC are protocols for the handling establishment and exchange of secure cryptography Keys called internet key

exchange protocol Ike igmp internet group management protocol protol the igmp protocol carries out the administration of Ip multicast group of end systems within a tcpip network special multicast rotors administer address list of end systems that can be addressed commonly via one multicast address OSF open shortest path first the OSF protocol a link state routing protocol transmits IP datagrams within a single routing domain or autonomous system OSF is the most widely used routing protocol on the internet third layer in TCP IP model is transport layer the transport layer establishes a basic data channel that an application uses in its task specific data exchange the layer establishes process to process connectivity by providing

endtoend services that are independent of the structure of user data and the logistics of exchanging in information for any particular specific purpose the protocol of the transport layer establishes a direct virtual endtoend communication connection to allow multiple application programs on the same computer every application program is assigned a port number to provide unique identification on the transport layer every unit of data sent must contain the port number of the sender and the receiver in order to be transmitted correctly together with the IP address the port number defines what is commonly known as Network socket a unique connection end point in the network work endtoend message transmission or

connecting applications at the transport layer can be categorized as either connection oriented which is implemented in TCP or connectionless implemented in UDP the transport control protocol TCP is a core element of the Internet Protocol architecture and the most popular protocol of the transport layer in the TCP IP reference model standardized as RFC 793 it carries out a reliable connection oriented bidirectional data exchange between two end systems TCP enables the establishment of virtual networks next the user data protocol UDP is the second most prominent protocol of the transport layer standardized as RFC 768 it transmits independent data units known as datagram between application programs that reside on different

computers in networks the transmission is unreliable possibly combined with data loss proliferation of datagrams and changes in sequence the datagrams recognized as false are discarded by UDP and do not even reach the receiver in comparison to TCP UDP is clearly less complex which is reflected in its increased data throughput but UDP suffers a dramatic loss of reliability and security other important Protocols of the transport layer are dccp that is datagram congestion control protocol which is a message oriented protocol of the transport layer in addition to reliably establishing and terminating connections it also distributes overload notifications it provides overload or congestion control and can be used for

the negotiation of transmission parameters RSVP that is resource reservation protocol which is used for the request and reservation of network resources using IP to transmit data streams it is not intended for the actual data transport and bears a similarity to the mcomp and igmp protocols at the internet layer RSVP can be implemented by end systems as well as routers without having to reserve and maintain specified service qualities TLS that is transport layer security is predecessor of the secure socket layer protocol SSL TLS supplies cryptographic protocols for secure data transport in Internet individual TCP segments are encrypted by TLS and 26 minutes, 1 second SSL TLS provides protocols for the negotiation of transmission parameters

for the exchange of cryptographic keys and authentication as well as for encryption and digital signature SCP that is stream control transmission protocol is a proposal for a highly scalable and performance version of the original TCP protocol which is reliable connection oriented and is specialized in transmitting large amounts of data next we move on to last layer of TCP IP model that is application layer the functions on the application layer in the TCP IP reference model can be mapped onto layer 5 to 7 in the iso OSI reference model the application layer primarily functions as an interface for the actual application programs wishing to communicate over the network a few of the many important

Protocols of the application layer located in the tcpip protocol family are tnet that is telnet Network which is defined as RFC 854 which allows the setup of an interactive bidirectional communication connection to a remote computer through a command line interface FTP that is file transfer protocol defined by RFC 959 facilitates the transmission and manipulation of

data between two computers connected over a tcpip network FTP functions according to the client server Paradigm a client initiates the connection and request a service the server takes the connection request and answers the service request SMTP that is simple mail transfer protocol defined by RFC 821 is a simple structured protocol for the transmission of electronic mail on

the internet HTTP that is hyper text transport protocol defined by RFC 2616 is used for data transmission in the world wide web just as many other Protocols of the application layer it works according to the client server Paradigm and is based on the reliable transport protocol TCP RPC that is remote procedure call defined by RFC 1057 and RFC 5531 is used for interprocess communication that allows a computer program to call an external sub routine located in another addressing area DNS that is domain name system establishes a name and directory service that delivers the assignment of readable and system names to IP addresses for all participating systems on the internet SNMP that is simple mail Network management protocol defined in

341 RFC helps Network Management Systems monitor administer and control individual ual systems connected to the network RTP that is realtime transport protocol defined by rfc1 189 is used for transmission of real time audio and video data over internet so friends this was about TCP IP reference model in this lecture we first discussed about introductory concepts of TCP IP reference model which was followed by layered protocol architecture of tcpip reference model in the end we discussed different protocols at different layers of this model thank you